

Fiber Manufacturing

Main features

Preparation of Optical Fibers

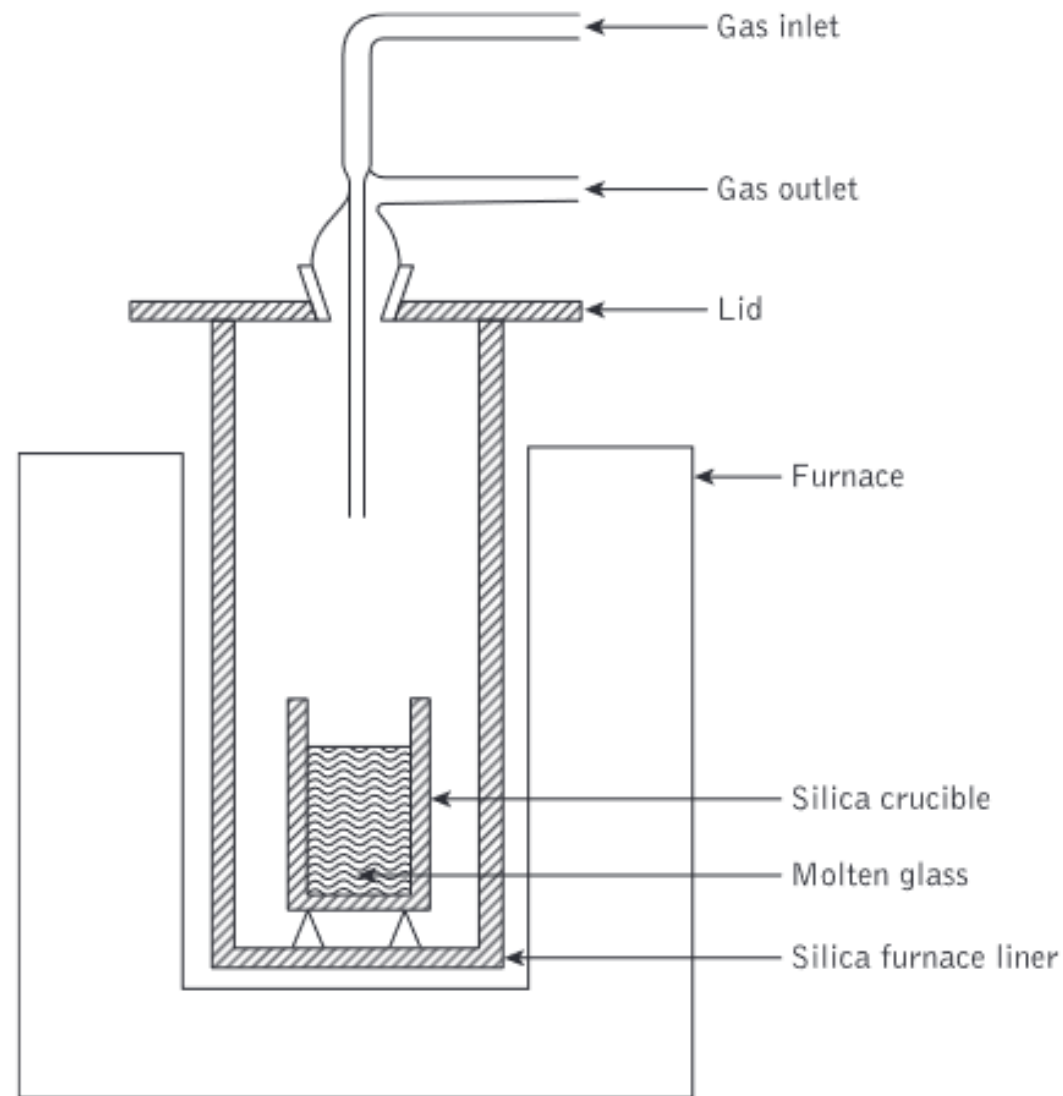


Figure 4.1 Glassmaking furnace for the production of high-purity glasses

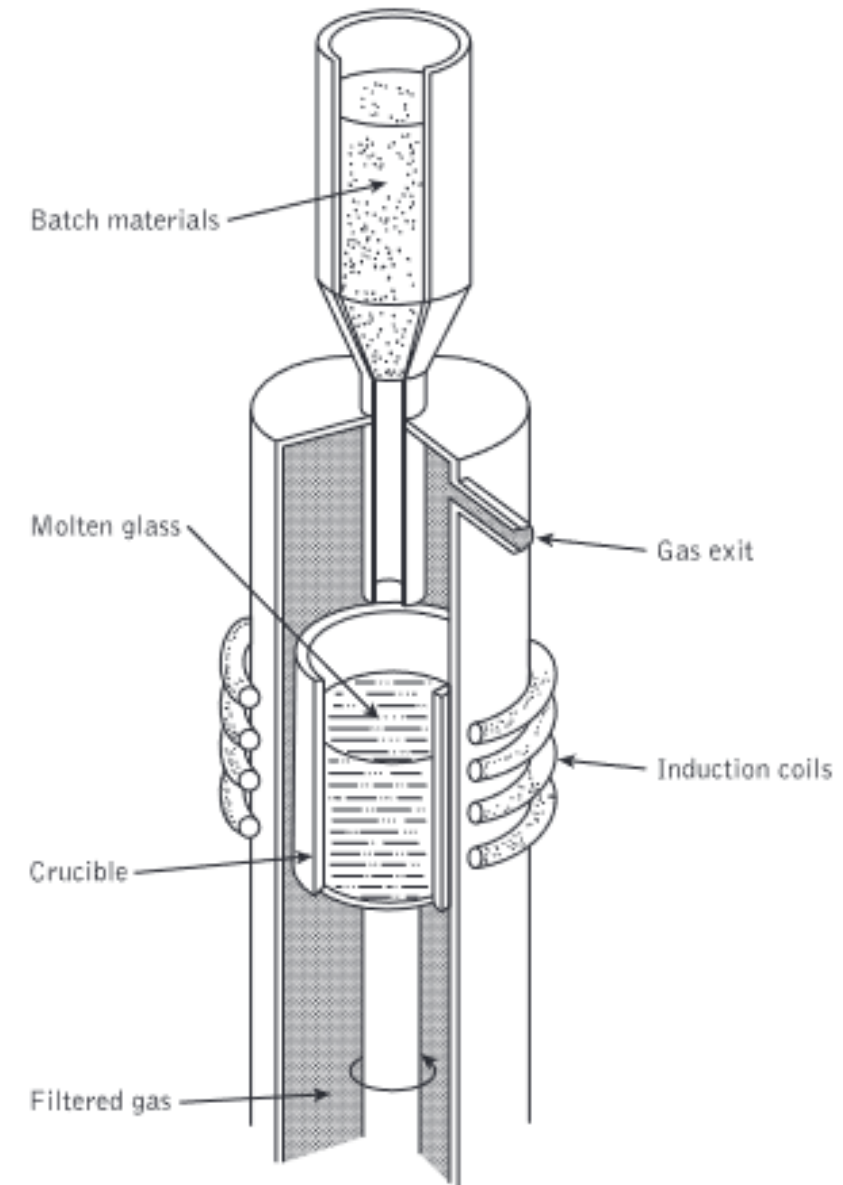


Figure 4.2 High-purity melting using a radio-frequency Induction furnace

How to produce the preform and materials involved

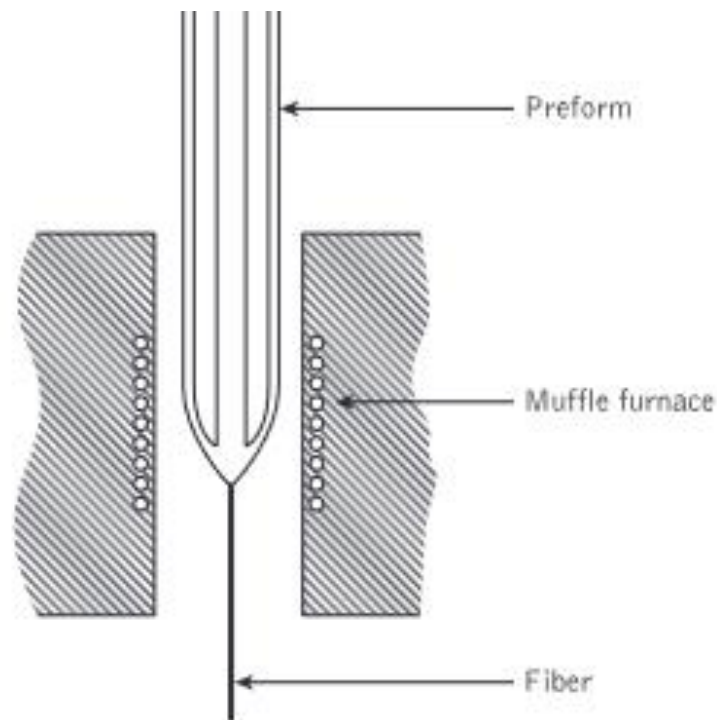
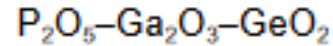
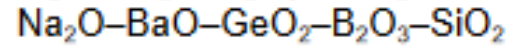
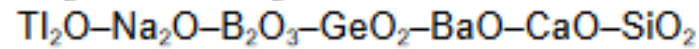
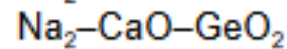
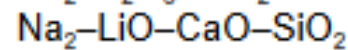
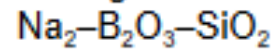


Figure 4.3 Optical fiber from a preform [Ref. 9]

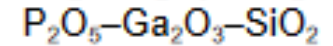
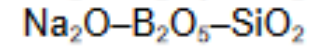
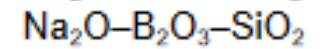
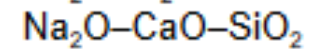
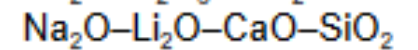
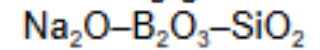
Table 4.1 Material systems used in the fabrication of multicomponent glass fibers by the double crucible technique

Step index

Core glass

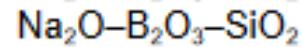
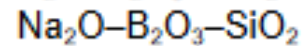
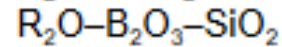
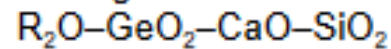


Cladding glass

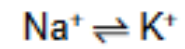


Graded index

Base glass



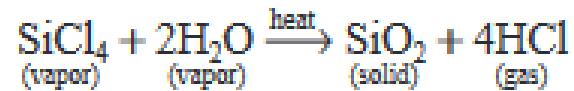
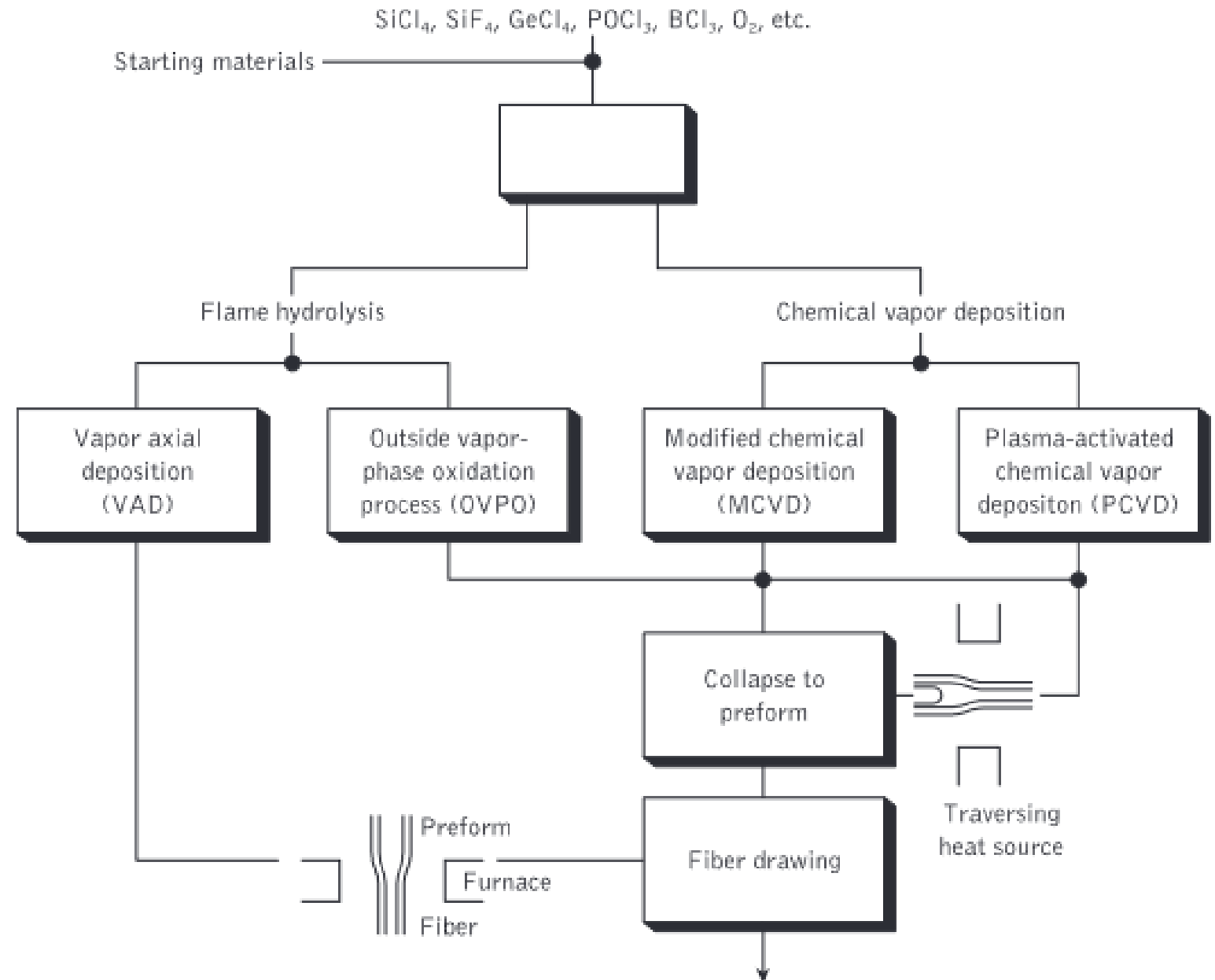
Diffusion mechanism



Na_2O diffusion

CaO , BaO , diffusion

Vapor Phase Techniques



and:



or:

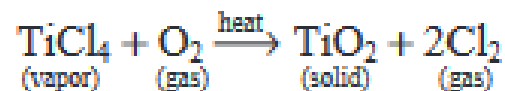


Figure 4.6 Schematic illustration of the vapor-phase deposition techniques used in the preparation of low-loss optical fibers

Refractive Index of Silica

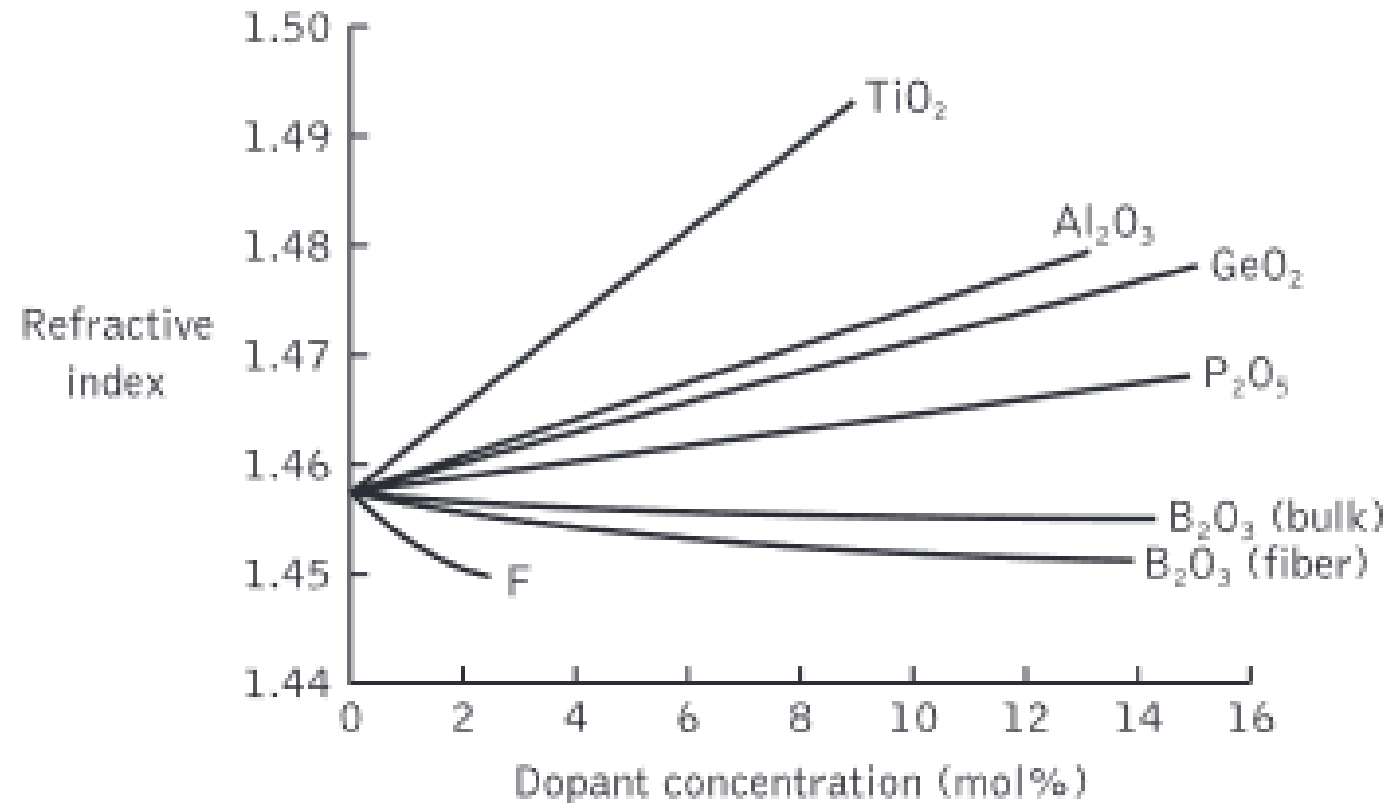


Figure 4.5 The variation in the refractive index of silica using various dopants. Reproduced with permission from the publishers, Society of Glass Technology, *Phys. Chem. Glasses*, **21**, p. 5, 1980

External Deposition

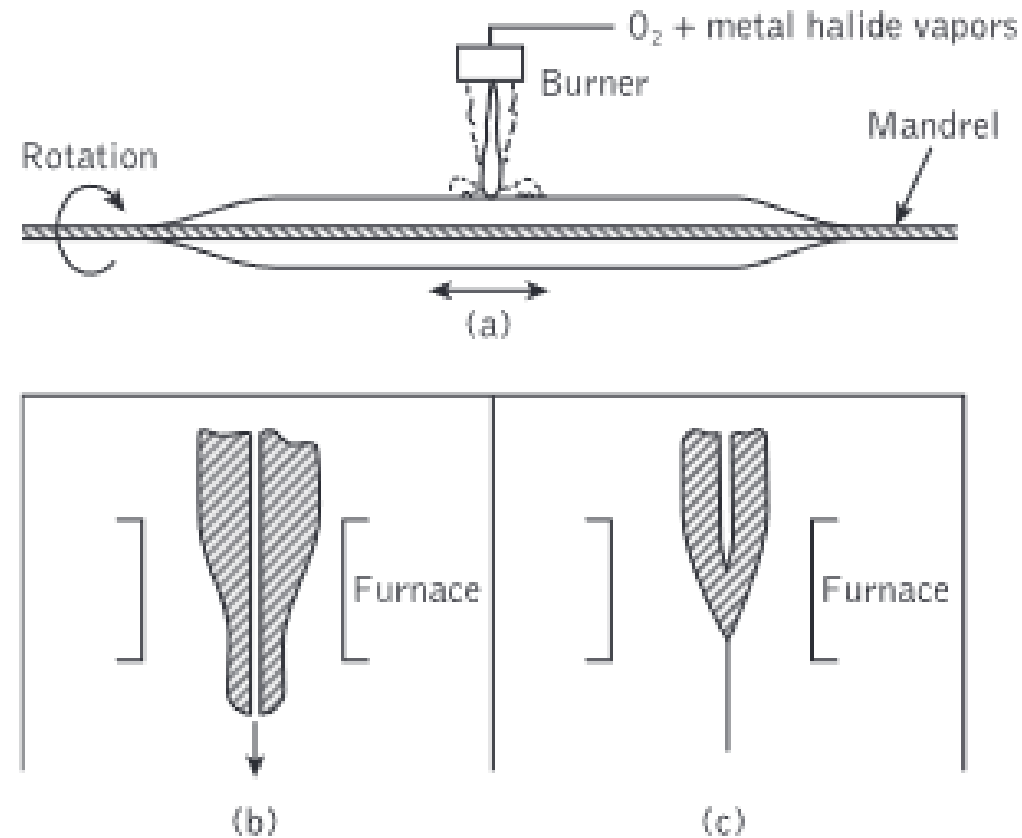
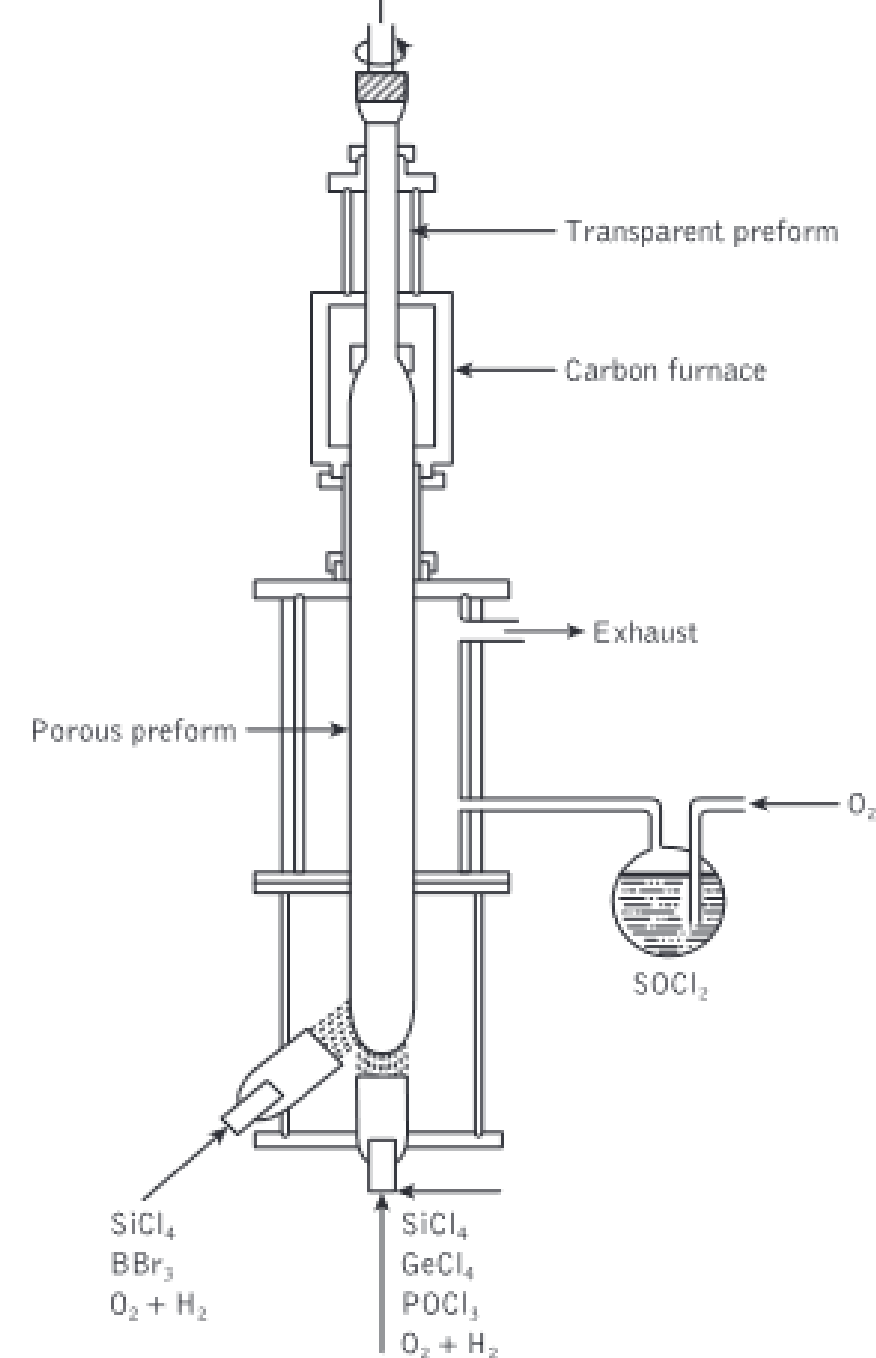
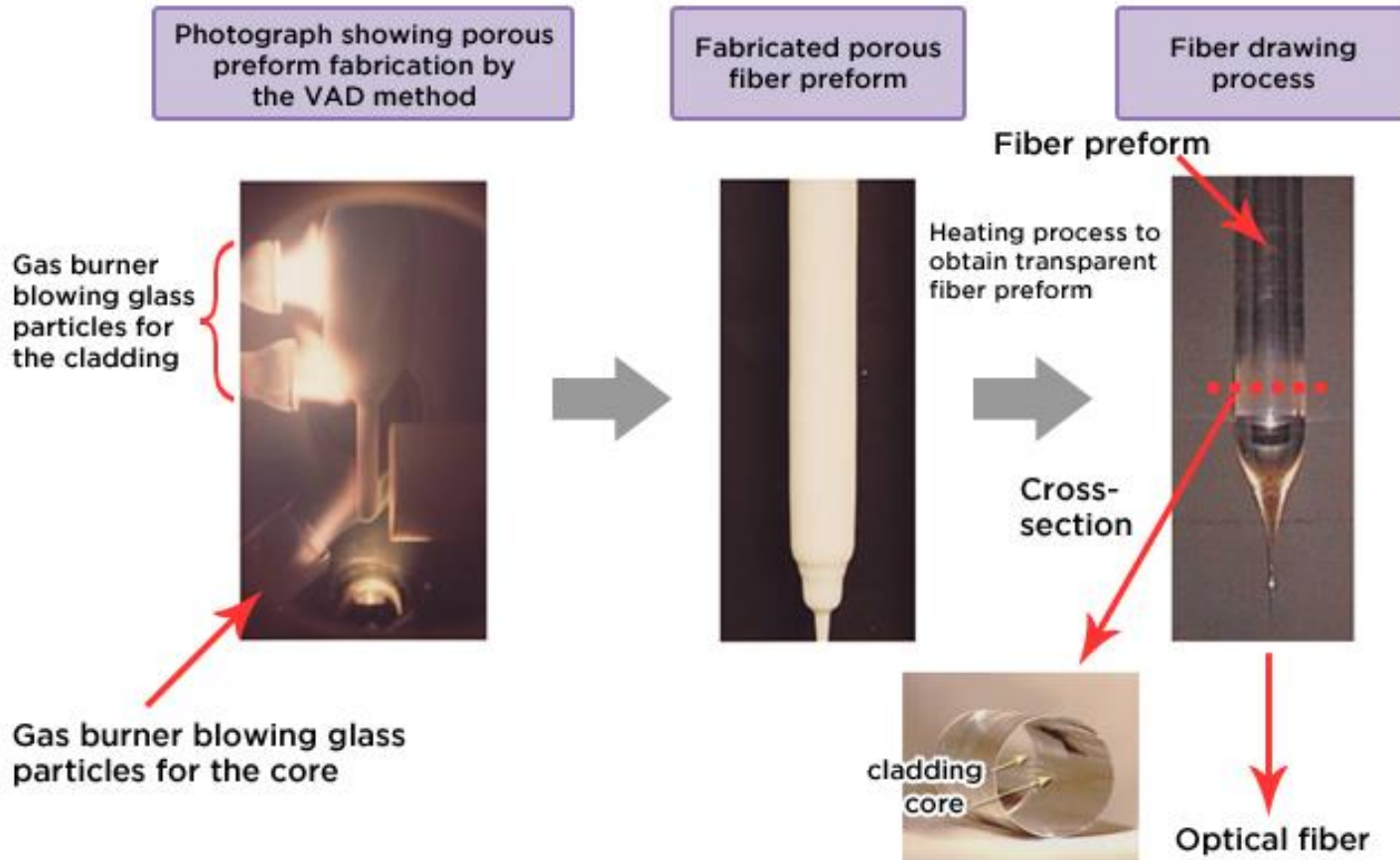


Figure 4.7 Schematic diagram of the OVPO process for the preparation of optical fibers: (a) soot deposition; (b) preform sintering; (c) fiber drawing. Reprinted from Ref. 17 with permission from Elsevier

Vapour Axial Deposition



Modified chemical vapor deposition

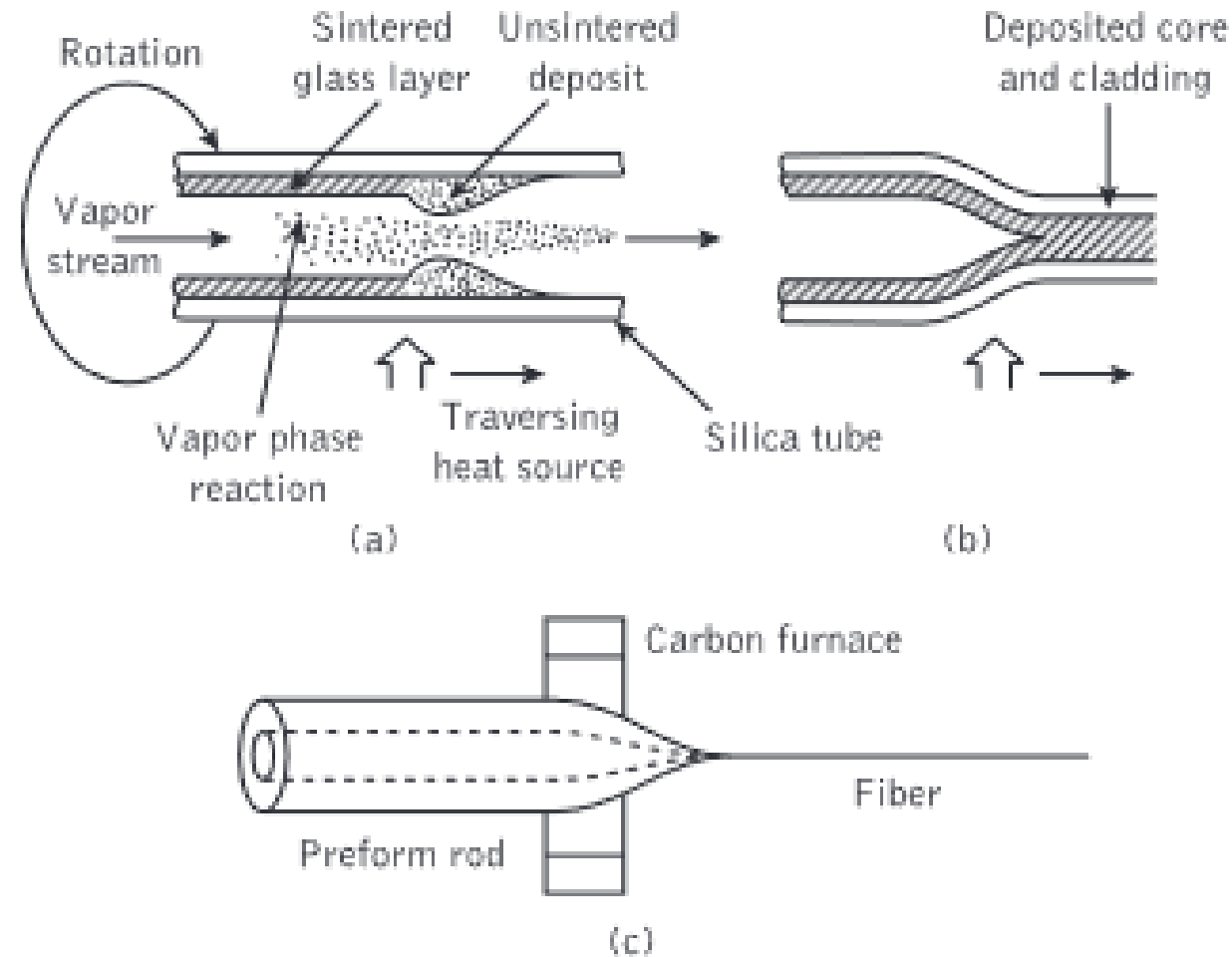


Figure 4.9 Schematic diagram showing the MCVD method for the preparation of optical fibers: (a) deposition; (b) collapse to produce a preform; (c) fiber drawing

A Fiber-Optic Link

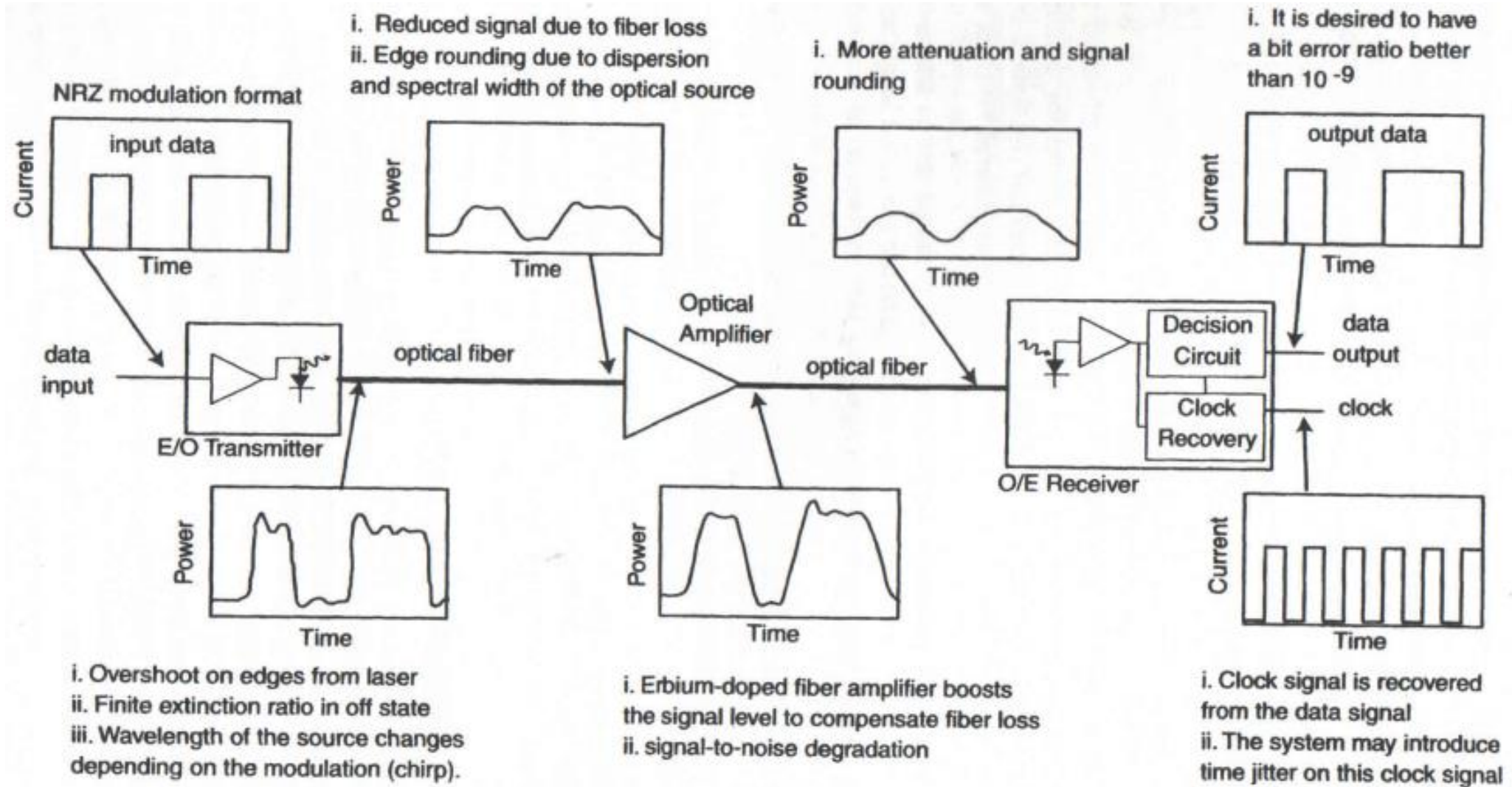


Figure 1.1 A digital fiber-optic communication link.